

Reflexive ideals

Value semigroups and good semigroups, Catania 2026

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Preliminaries

This is a joint work with - P. Campochiaro, and - M. D'Anna.

It is inspired by

- the (unpublished) work of Barucci, a
- the master thesis of P. Campochiaro (M. D'Anna as advisor), and
- the work of D. Spirito and his coauthors on star operations in numerical semigroups.

It was motivated by the idea of irreducibility with respect to the operation of intersection.

Ideals of numerical semigroups

A numerical semigroup is a subset of the non-negative integers that is closed under addition, contains zero, and has a finite complement in the non-negative integers.

An ideal of a numerical semigroup S is a non-empty subset I of the non-negative integers such that

- $I + S \subseteq I$, and
- there exists $z \in \mathbb{Z}$ such that $z + I \subseteq S$.

If I and J are ideals of S , the following sets are also ideals of S :

- $I + J = \{i + j : i \in I, j \in J\}$,
- $z + I = (z + S) + I$ (ideals of the form $z + S$ are called principal ideals),
- $I \cup J$,
- $I \cap J$,
- $I - J = \{z \in \mathbb{Z} : z + J \subseteq I\}$ (quotient).

Properties of quotients

Given I, J and K ideals of S and $x \in \mathbb{Z}$, the following properties hold:

- $(x + I) - J = x + (I - J)$
- $I - (x + J) = -x + (I - J)$;
- if $J \subseteq K$, then
 - $I - J \supseteq I - K$, and
 - $J - I \subseteq K - I$;
- $I - (J \cap K) \supseteq (I - J) \cup (I - K)$;
- $I - (J \cup K) = (I - J) \cap (I - K)$;
- $(I - J) - K = I - (J + K)$;
- $I + J \subseteq K$ if and only if $I \subseteq K - J$.

It follows that $J - I = J - (J - (J - I))$.

Dual and reflexive closure

For an ideal I of S , we define the *dual* of I as

$$I^* = S - I.$$

The *reflexive closure* of I is defined as

$$I^\vee = (I^*)^* = S - (S - I).$$

Given I and J ideals of S , and a an integer

- if $I \subseteq J$, then $I^\vee \subseteq J^\vee$;
- $I \subseteq I^\vee$;
- $(I^\vee)^\vee = I^\vee$;
- $I^\vee \cup J^\vee \subseteq (I \cup J)^\vee$;
- $(I \cap J)^\vee \subseteq I^\vee \cap J^\vee$;
- $(a + I)^\vee = a + I^\vee$.

The operation $I \mapsto I^\vee$ is a star operation

Reflexive closure and principal ideals

The order associated to a numerical semigroup S is defined as

$$x \leq_S y \text{ if and only if } y - x \in S.$$

Every ideal of S is a union of principal ideals:

$$I = \text{Minimals}_{\leq_S}(I) + S = \bigcup_{x \in \text{Minimals}_{\leq_S}(I)} (x + S).$$

The following is a refinement of an unpublished result of Barucci:

$$I^\vee = \bigcap_{z \in A} -z + S,$$

where $A = \text{Minimals}_{\leq_S}(S - I)$.

 This decomposition is not minimal nor unique

Let $S = \langle 6, 7, 16 \rangle$ and let $I = \{0, 1, 3, 5\} + S$. Then,

$$\text{Minimals}_{\leq_S}(S - I) = \{13, 18, 21, 22, 23\},$$

while $I = (-13 + S) \cap (-18 + S) \cap (-21 + S)$.

Reflexive closure and canonical ideal

Let S be a numerical semigroup. Recall that the *canonical ideal* of S is defined as

$$K(S) = \{x \in \mathbb{Z} : f(S) - x \notin S\},$$

where $f(S)$ is the Frobenius number of S , that is, $\max(\mathbb{Z} \setminus S)$.

It is known (Jäger) that for every ideal I of S , we have that

$$K(S) - (K(S) - I) = I$$

For every ideal I of S , we have that

$$I^\vee = (K(S) + I) - K(S).$$

Thus,

$$I \subseteq I^\vee \subseteq I + K(S).$$

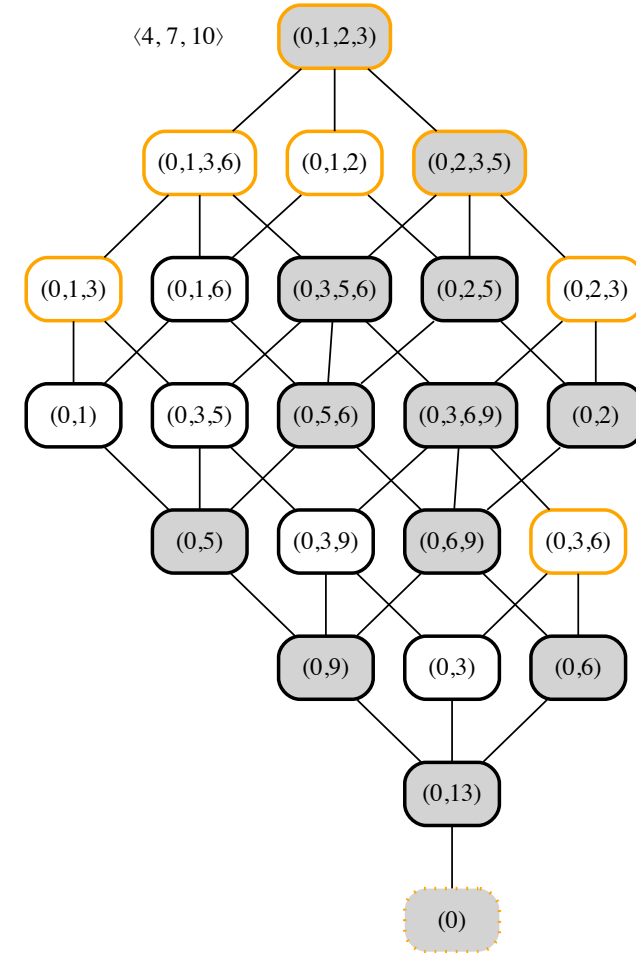
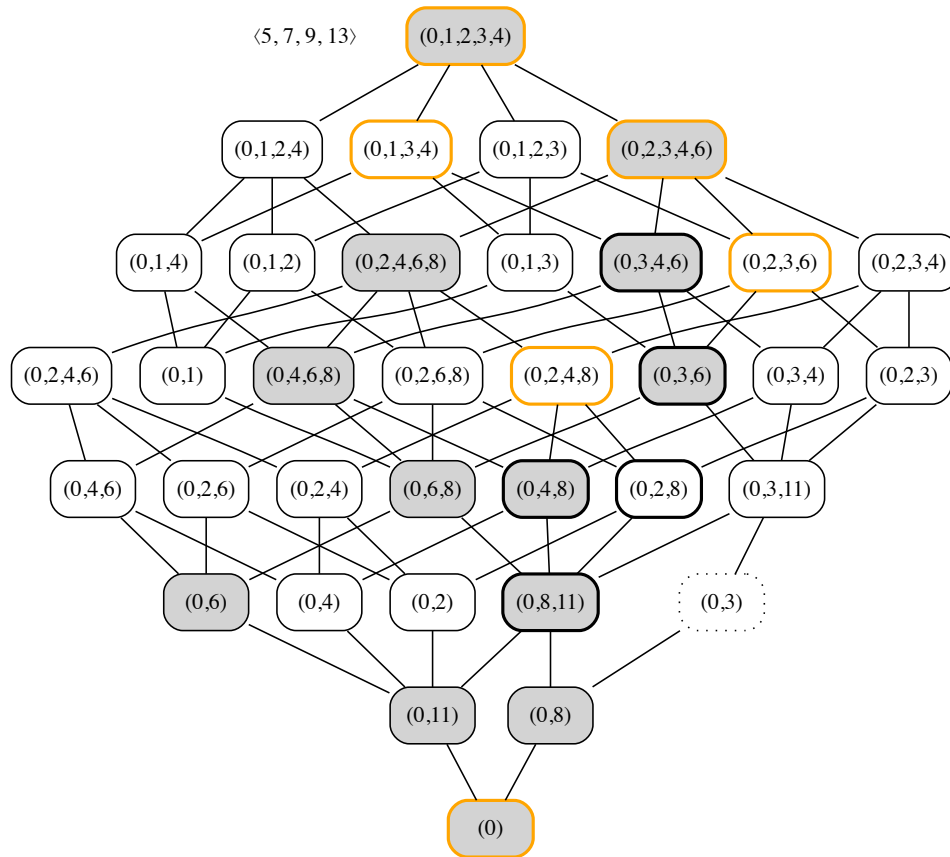
Reflexive ideals

Let S be a numerical semigroup and let I be an ideal of S .

We say that I is *reflexive* if $I = I^\vee$.

- Being reflexive is closed under intersection and translation.
- Thus, one could focus on normalized ideals, that is, ideals I such that $\min(I) = 0$.
- The set of normalized ideals of S is isomorphic to the ideal class monoid of S , and has finitely many elements.
- Every normalized reflexive ideal is a finite intersection of *irreducible* reflexive ideals, that is, reflexive ideals that cannot be expressed as the intersection of two reflexive ideals properly containing it, which we call R-irreducible ideals.
- By knowing the R-irreducible ideals of S , we can compute all reflexive ideals of S up to translation.

Examples



In gray idempotent ideals (over-semigroups), bold reflexive ideals, orange R-irreducible ideals, and dotted the canonical ideal.

Integrally closed ideals

Let S be a numerical semigroup and let $s \in S$. The set $S(s) = \{t \in S : t \geq s\}$ is an ideal of S .

We say that an ideal I of S is if $I = S(s)$ for some $s \in S$.

i Integrally closed ideals are reflexive

If I is the translation of an integrally closed ideal, then I is reflexive.

For I and ideal of S , we define the *conductor* of I as $c(I) = f(I) + 1$, which is the smallest integer c such that $c + \mathbb{N} \subseteq I$.

We define $\tilde{I} = c(S) - (c(S) - I)$.

i Integrally closed ideals are R-irreducible

If I is a normalized ideal and $\tilde{I}$ is integrally closed, then I is R-irreducible.

Decomposition into translates of integrally closed ideals

Let S be a numerical semigroup and let $M = S \setminus \{0\}$ be its maximal ideal.

Theorem 1 If I is a normalized reflexive ideal of S , then there exist an anti-chain with respect to the order induced by $M - M$, $s_1, \dots, s_n \in S$ (minimal generators of $S - I$), such that

$$I = (-s_1 + S(s_1)) \cap \dots \cap (-s_n + S(s_n)).$$

As a consequence of this,

I is R -irreducible if and only if $\tilde{I}$ is integrally closed.

Computing all normalized reflexive ideals of S

This also provides a procedure to compute the set of reflexive normalized ideals of a numerical semigroup S . We start by computing $M - M$, with M the maximal ideal of S . Then, we compute the set of all anti-chains of small elements of S with respect to the order induced by $M - M$. For each anti-chain A , we compute $\cap_{x \in A} -x + S(x)$.

This procedure is faster than computing the set of all normalized ideals and then filter those that are reflexive.

Lengths of these decompositions

Let S be a numerical semigroup and let I be an ideal of S .

The *type* of I is defined as $t(I) = |(I - M) \setminus I|$, where M is the maximal ideal of S

Conjecture

Let I be a normalized reflexive ideal of S . Then I admits a decomposition into R -irreducible ideals of length at most $t(I)$.

This conjecture has been tested for numerical semigroups with Frobenius number up to 25 (with NumericalSgps).

Conjecture

Sets of lengths of decompositions are always intervals (of size at most two).

Apéry sets

Let S be a numerical semigroup and let I be an ideal of S

Let m be a non-zero element of S (usually the multiplicity of S - the minimum of its maximal ideal)

The Apéry set of I with respect to m is

$$\text{Ap}(I, m) = \{i \in I : i - m \notin I\}.$$

This set has m elements, $\{a_0, a_1, \dots, a_{m-1}\}$, where a_i is the minimum element in I congruent with i modulo m .

If $\text{Ap}(I, m) = \{a_0, \dots, a_{m-1}\}$ and $\text{Ap}(J, m) = \{b_0, \dots, b_{m-1}\}$, then

- $I \subseteq J$ if and only if $b_i \leq a_i$ for all i
- $\text{Ap}(I \cap J, m) = \{\max\{a_0, b_0\}, \dots, \max\{a_{m-1}, b_{m-1}\}\}$
- $\text{Ap}(I \cup J, m) = \{\min\{a_0, b_0\}, \dots, \min\{a_{m-1}, b_{m-1}\}\}$

Example

Let $S = \langle 8, 9, 10, 11, 12, 15 \rangle$, $I = \{0, 7, 13, 14\} + S$.

Then $M - M = S \cup \{7, 14\}$ and $S - I = \{8, 9, 10, 11, 12, 15\} + S$.

We have that $8 \leq_{M-M} 15$, and so 15 is not needed for the representation of I as the intersection of translations of integrally closed ideals.

If we list the Apéry sets of $-s_i + S(s_i)$ for the remaining minimal generators of $S - I$, we get:

$$\begin{aligned} \text{Ap}(S(8) - 8, 8) &= \{0, 1, 2, 3, 4, 13, 14, 7\}, & \text{Ap}(S(11) - 11, 8) &= \{0, 1, 10, 11, 4, 5, 6, 7\}, \\ \text{Ap}(S(9) - 9, 8) &= \{0, 1, 2, 3, 12, 13, 6, 7\}, & \text{Ap}(S(12) - 12, 8) &= \{0, 9, 10, 3, 4, 5, 6, 7\}, \\ \text{Ap}(S(10) - 10, 8) &= \{0, 1, 2, 11, 12, 5, 6, 7\}. \end{aligned}$$

Then $I = (-8 + S(8)) \cap (-10 + S(10)) \cap (-12 + S(12))$ is a non-redundant factorization of I .

	0	1	2	3	4	5	6	7
$\text{Ap}(-8 + S(8), 8)$	0	1	2	3	4	13	14	7
$\text{Ap}(-10 + S(10), 8)$	0	1	2	11	12	5	6	7
$\text{Ap}(-12 + S(12), 8)$	0	9	10	3	4	5	6	7
$\text{Ap}(I, 8)$	0	9	10	11	12	13	14	7

Apéry set of a quotient

Let S be a numerical semigroup and let I and J ideals of S .

Let $\text{Ap}(I, m) = \{a_0, a_1, \dots, a_{m-1}\}$ and $\text{Ap}(J, m) = \{b_0, b_1, \dots, b_{m-1}\}$. Then,

$\text{Ap}(J - I, m) = \{x_0, x_1, \dots, x_{m-1}\}$, where

$$x_i = \max\{b_{i+j} - a_j : j \in \{0, \dots, m-1\}\}.$$

In particular, $\text{Ap}(I^\vee, m) = \{y_0, y_1, \dots, y_{m-1}\}$, with

$$y_i = \max\{w_{i+j} + \min\{a_k - w_{j+k} : k \in \{0, \dots, m-1\}\} : j \in \{0, \dots, m-1\}\}.$$

Reflexive ideals in some families of numerical semigroups

Let S be a numerical semigroup and let I be an ideal of S .

Recall that $I \subseteq I^\vee \subseteq I + K(S)$.

let T be an over-semigroup of S . The following are equivalent (Barucci).

- $K(S) \subseteq T$.
- Every ideal of T is a reflexive ideal of S .

Characterization

The following are equivalent

- S is symmetric ($K(S) = S$).
- Every ideal of S is reflexive.

Almost symmetric

Let S be a numerical semigroup with Frobenius number f , genus g (number of gaps), and type t .

We say that S is *almost symmetric* if $g = (f + t)/2$.

Every irreducible numerical semigroup (cannot be expressed as the intersection of two proper over-semigroups) is almost symmetric.

If K is the canonical ideal of S and M is the maximal ideal of S , then S is almost symmetric if and only if $K \subseteq M - M$

Characterization

The following are equivalent.

- S is almost symmetric.
- A non-principal ideal I of S is reflexive if and only if $K \subseteq I - I$
- A non-principal ideal I of S is reflexive if and only if I is an ideal of $M - M$.

Maximal embedding dimension

Let S be a numerical semigroup and let M be its maximal ideal.

It is well known that the minimal generating set of S is $M \setminus (M + M)$. The size of this set is the *embedding dimension* of S and it is always less than or equal to the multiplicity of S .

We say that S has *maximal embedding dimension* if the embedding dimension of S equals its multiplicity.

Characterization

The following are equivalent.

- S has maximal embedding dimension.
- A non-principal ideal I is reflexive if and only if I is a reflexive ideal of $M - M$.
- M is $(M - M)$ -reflexive ($M = (M - M) - ((M - M) - M)$).

Arf numerical semigroups

Let S be a numerical semigroup.

We say that S is *Arf* if for every $x, y, z \in S$ with $x \geq y \geq z$ we have $x + y - z \in S$, equivalently, every integrally closed ideal I of S is stable ($I + I = \min(I) + I$).

Characterization

The following are equivalent.

- S is Arf.
- A non-principal ideal I is reflexive if and only if I is the translate of an integrally closed ideal.
- A normalized ideal is reflexive if and only if it is R-irreducible.

Thank you for your attention!

More info at numerical-semigroups.github.io.